

**FOURTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

MTS 4B 04—LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 4B 04—LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. If A and B are square matrices of the same order, then $\text{tr}(AB) =$:
 - (A) $\text{tr}(A + B)$.
 - (B) $\text{tr}(A) \text{tr}(B)$.
 - (C) $\text{tr}(BA)$.
 - (D) $\text{tr}(A) + \text{tr}(B)$.
2. If A and B are symmetric matrices of same order, then :
 - (A) AB is always symmetric.
 - (B) AB is never symmetric.
 - (C) AB is skew-symmetric.
 - (D) AB is symmetric if and only if $AB = BA$.
3. A matrix that is both symmetric and upper triangular must be a :
 - (A) Diagonal matrix.
 - (B) Non-diagonal but symmetric.
 - (C) Both (A) and (B).
 - (D) None of the above.
4. A matrix E is called _____ if it can be obtained from an identity matrix by performing a single elementary row operation.
 - (A) Equivalent matrix.
 - (B) Echelon matrix.
 - (C) Elementary matrix.
 - (D) Row reduced matrix.
5. A homogeneous linear system in n unknowns whose corresponding augmented matrix has a reduced row echelon form with r leading 1's has :
 - (A) n free variables.
 - (B) $n - r$ free variables.
 - (C) r free variables.
 - (D) Cannot be determined.
6. If A is an $n \times n$ matrix that is not invertible, then the linear system $Ax = 0$ has :
 - (A) Infinitely many solutions.
 - (B) Exactly one solution.
 - (C) Not possible to find solution.
 - (D) Finitely many solutions.
7. If A is an $m \times n$ matrix, then the codomain of the transformation T_A is :
 - (A) $\mathbb{R}^n$.
 - (B) $\mathbb{R}^{m+n}$.
 - (C) $\mathbb{R}^{mn}$.
 - (D) $\mathbb{R}^m$.

8. If $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and if $T_A(x) = 0$ for every vector x in $\mathbb{R}^n$, then A is :
- (A) The $n \times n$ zero matrix. (B) The $n \times n$ identity matrix.
 (C) An elementary matrix. (D) Cannot be determined.
9. Which of the following is false ?
- (A) Every subspace of a vector space is itself a vector space.
 (B) Every vector space is a subspace of itself.
 (C) The intersection of any *two* subspaces of a vector space V is a subspace of V .
 (D) The union of any *two* subspaces of a vector space V is a subspace of V .
10. The polynomials $x - 1, (x - 1)^2, (x - 1)^3$ span P^3 .
- (A) True. (B) False.
 (C) Data not complete. (D) Span P^4 .
11. The kernel of a matrix transformation $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a subspace of :
- (A) $\mathbb{R}^n$. (B) $\mathbb{R}^m$.
 (C) $\mathbb{R}^{n+m}$. (D) $\mathbb{R}^{nm}$.
12. The dimension of zero vector space is :
- (A) Not defined. (B) 1.
 (C) 0. (D) Infinite.
13. Which of the following is true ?
- (A) Every linearly independent set of five vectors in $\mathbb{R}^5$ is a basis for $\mathbb{R}^5$.
 (B) Every set of five vectors that spans $\mathbb{R}^5$ is a basis for $\mathbb{R}^5$.
 (C) Every set of vectors that spans $\mathbb{R}^5$ contains a basis for $\mathbb{R}^5$.
 (D) All are true.
14. Which of the following is not a vector space ?
- (A) The set of all 2×2 invertible matrices with the standard matrix addition and scalar multiplication.
 (B) The set of all 2×2 matrices of the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with the standard matrix addition and scalar multiplication.
 (C) The set of all 2×2 matrices with real entries with the standard matrix addition and scalar multiplication.
 (D) None of these.

Turn over

15. Let A be any matrix. Then :

- (A) $\text{rank}(A) = \text{rank}(A^T)$. (B) $\text{rank}(A) \neq \text{rank}(A^T)$.
 (C) $\text{rank}(A) < \text{rank}(A^T)$. (D) $\text{rank}(A) > \text{rank}(A^T)$.

16. If W is a subspace of $\mathbb{R}^n$, then which of the following statement is false.

- (A) $W^\perp$ is a subspace of $\mathbb{R}^n$.
 (B) The only vector common to W and $W^\perp$ is 0.
 (C) The orthogonal complement of $W^\perp$ is W .
 (D) None of these.

17. If A is an $m \times n$ matrix, then :

- (A) The null space of A and the row space of A are orthogonal complements in $\mathbb{R}^n$.
 (B) The null space of A^T and the column space of A are orthogonal complements in $\mathbb{R}^m$.
 (C) Both (A) and (B) are correct.
 (D) Neither (A) nor (B) are correct.

18. If A is a 3×5 matrix, then the rank of A^T is at most :

- (A) 4. (B) 2.
 (C) 5. (D) 3.

19. If A is a 3×5 matrix, then the nullity of A^T is at most :

- (A) 4. (B) 2.
 (C) 5. (D) 3.

20. The values of r and s for which $\begin{bmatrix} 1 & 0 & 0 \\ 0 & r-2 & 2 \\ 0 & s-1 & r+2 \\ 0 & 0 & 3 \end{bmatrix}$ has rank 2 ?

- (A) $r = 2, s = 2$. (B) $r = 2, s = 1$.
 (C) $r = -2, s = 1$. (D) Cannot have rank 2.

FOURTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021

Mathematics

MTS 4B 04—LINEAR ALGEBRA

Time : Two Hours and a Half

Maximum : 80 Marks

Section A (Short Answer Type Questions)

Answer at least ten questions.

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 30.

1. Describe different possibilities for solution (x, y) of a system linear equations in the xy plane.
 What are consistent system ?

2. Suppose that the augmented matrix for a linear system has been reduced to the row echelon form

$$\text{as } \begin{bmatrix} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix} \text{ solve the system.}$$

3. Define trace of a square matrix. Find the trace of the matrix $A = \begin{bmatrix} -1 & 2 & 7 & 0 \\ 3 & 5 & -8 & 4 \\ 1 & 2 & 7 & -3 \\ 4 & -2 & 1 & 0 \end{bmatrix}$.

4. Show that the standard unit vectors

$$e_1 = (1, 0, \dots, 0), e_2 = (0, 1, 0, \dots, 0), e_3 = (0, 0, 1, 0, \dots, 0), \dots, e_n = (0, 0, \dots, 1) \text{ span } \mathbb{R}^n.$$

5. Find the co-ordinate vector of $w = (1, 0)$ relative to the basis $s = [\bar{u}_1, \bar{u}_2]$ of $\mathbb{R}^2$, where $\bar{u}_1 = (1, -1)$ and $\bar{u}_2 = (1, 1)$.

6. Write two important facts about the vectors in a finite dimensional vector space V .

Turn over

7. Consider the bases $B = [\bar{u}_1, \bar{u}_2]$ and $B' = [\bar{u}_1', \bar{u}_2']$ where

$\bar{u}_1 = (1, 0)$, $\bar{u}_2 = (0, 1)$, $\bar{u}_1' = (1, 1)$, $\bar{u}_2' = (2, 1)$. Find the transition matrix $P_{B' \rightarrow B}$ from B' to B .

8. Define row spaces and null spaces an $m \times n$ matrix.

9. If $R = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is the row reduced echelon form of a 3×3 matrix A , then verify the rank-

nullity formula.

10. Show that the operator $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates vectors through an angle θ is one-one.

11. Find the image of the line $y = 4x$ under multiplication by the matrix $A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$.

12. Confirm by multiplication that x is an eigen vector of A and find the corresponding eigen value if

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

13. Let A be an $n \times n$ matrix. Define inner product on $\mathbb{R}^n$ generated by A . Also write the generating matrix of the weighted Euclidean inner product $\langle u, v \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \dots + w_n u_n v_n$.

14. If u, v are vectors in a real inner product space V , then show that $\|u + v\| \leq \|u\| + \|v\|$.

15. If A is an $n \times n$ orthogonal matrix, then show that $\|Ax\| = \|x\|$ for all x in $\mathbb{R}^n$.

(10 × 3 = 30 marks)

Section B (Paragraph/Problem Type Questions)

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

16. Describe Column Row Expansion method for finding the product AB for two matrices A and B . Use this to find the product $AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \\ -3 & 5 & 1 \end{bmatrix}$.
17. If A is an invertible matrix, then show that A^T is also invertible and $(AT)^{-1} = (A^{-1})^T$.
18. Consider the vectors $u = (1, 2, -1)$ and $v = (6, 4, 2)$ in $\mathbb{R}^3$. Show that $w = (9, 2, 7)$ is a linear combination of u and v and that $w' = (4, -1, 8)$ is not a linear combination of u and v .
19. If $s = \{v_1, v_2, \dots, v_n\}$ is a basis for a vector space V , then show that every vector v in V can be expressed in form $v = c_1v_1 + c_2v_2 + \dots + c_nv_n$ in exactly one way. What are the co-ordinates of v relative to the basis s .
20. If A is a matrix with n columns, then define rank of A and show that $\text{rank}(A) + \text{nullity}(A) = n$.
21. Find the standard matrix for the operator $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that first rotates a vector counter clockwise about z -axis through an angle θ , then reflects the resulting vector about yz plane and then projects that vector orthogonally onto the xy plane.
22. Define eigen space corresponding to an eigen value λ of a square matrix A . Also find eigen value and bases for the eigen space of the matrix $A = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$.
23. If w is a sub-space of real inner product space v , then show that :
 - (a) $w^\perp$ is subspace of v .
 - (b) $w \cap w^\perp = \{0\}$.

(5 × 6 = 30 marks)

Turn over

Section C (Essay Type Questions)

Answer any **two** questions.

Each question carries 10 marks.

24. (a) Show that every elementary matrix is invertible and the inverse is also an elementary matrix.

(b) Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ using Row operations.

25. (a) Let V be a vector space and $\bar{u}$ a vector in V and K a scalar. Then show that :

(a) $0\bar{u} = 0$; and

(b) $(-1)\bar{u} = -\bar{u}$.

- (b) Show that the vectors $v_1 = (1, 2, 1)$, $v_2 = (2, 9, 0)$ and $v_3 = (3, 3, 4)$ form a basis for $\mathbb{R}^3$.

26. (a) Consider the basis $B = [u_1, u_2]$ and $B' = [u_1', u_2']$ for $\mathbb{R}^2$ where $u_1 = (2, 2)$, $u_2 = (4, -1)$

$$u_1' = (1, 3), u_2' = (-1, -1)$$

- (i) Find the transition matrix B' to B .

- (ii) Find the transition matrix B to B' .

- (b) Find the reflection of the vector $x = (1, 5)$ about the line through the origin that makes an

angle of $\frac{\pi}{6}$ with the x -axis.

27. When you can say that a square matrix A is diagonalizable ? If A is an $n \times n$ matrix, show that the following statements are equivalent :

- (a) A is diagonalizable ; and

- (b) A has n linearly independent eigen vectors.

(2 × 10 = 20 marks)